



Course Project Description

CS 6341 Robotics

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The University of Texas at Dallas

Robot Hardware: SO101 Arm



<https://github.com/TheRobotStudio/SO-ARM100>

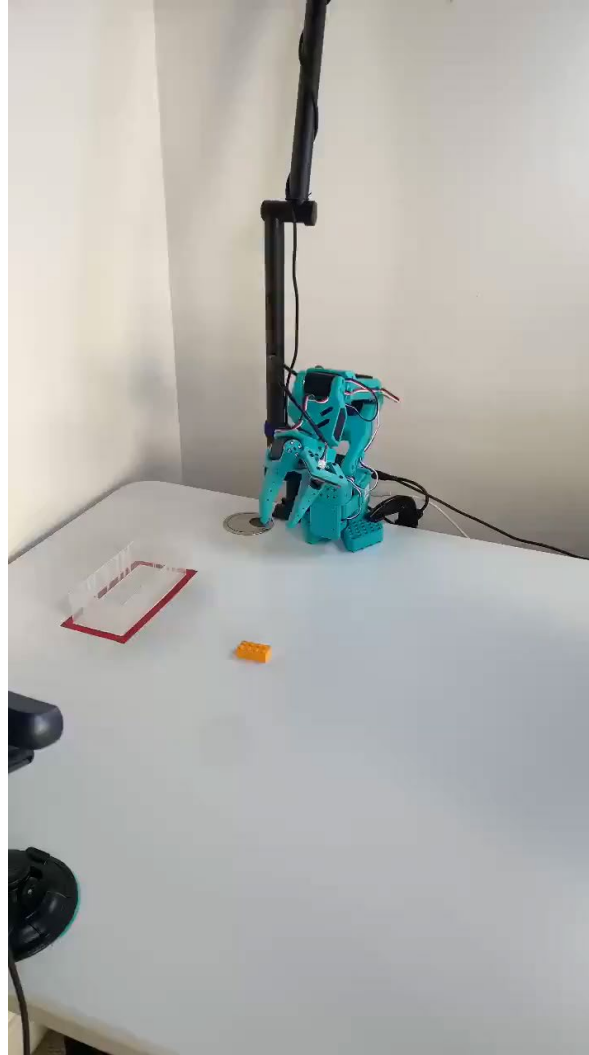
Course Project

- Team Project (45%)
 - 3 or 4 students for a project
 - Project proposal (10%)
 - Project mid-term report (10%)
 - Project presentation (15%): **in-class demo with the SO101 arm**
 - Project final report (10%)

Course Project Tracks

- Research-oriented
 - Proposal a new idea in robotics that has not been explored before
 - Implement the new idea and conduct experiments to verify it
- Application-oriented
 - Apply an existing algorithm or method to a new problem or a new application
 - E.g., if a method is proposed for domain A, explore applying it to a different domain
- Implementation-oriented
 - Select an existing algorithm or method, implement it and conduct experiments to verify the implementation
 - Cannot just use open-source code and run experiments with it

Example: Pick-and-Place



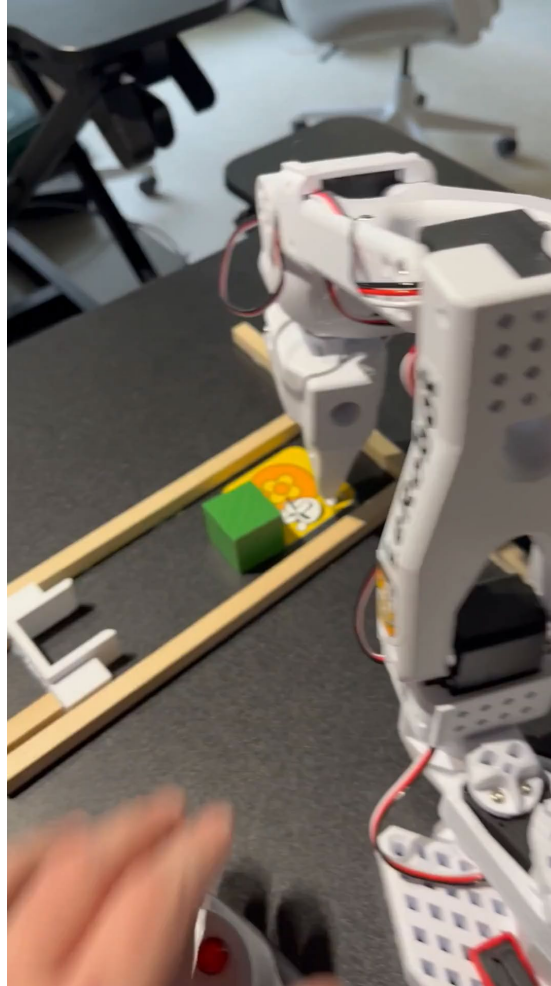
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Example: Pick-and-Place



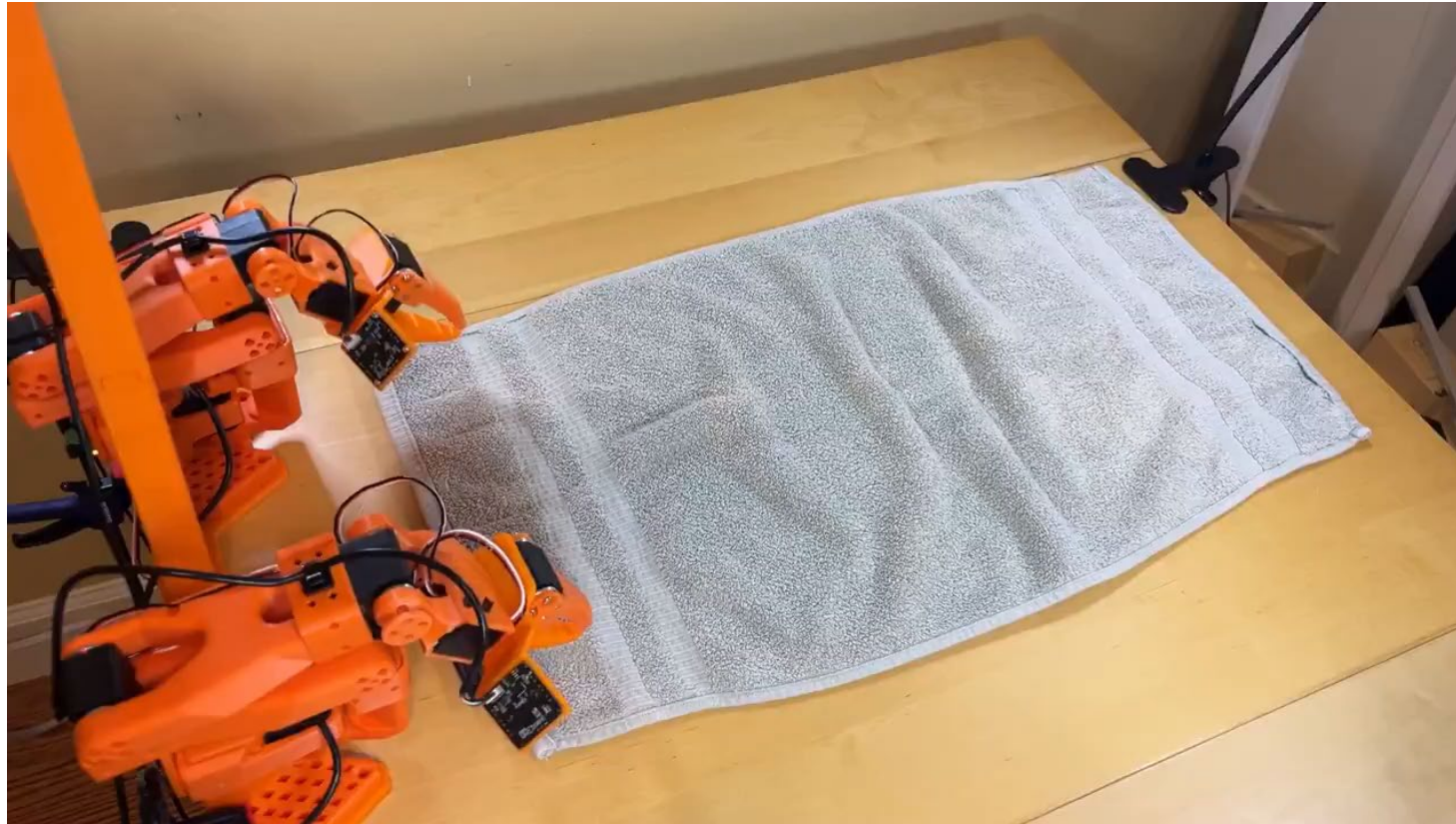
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Example: Robot Pushing



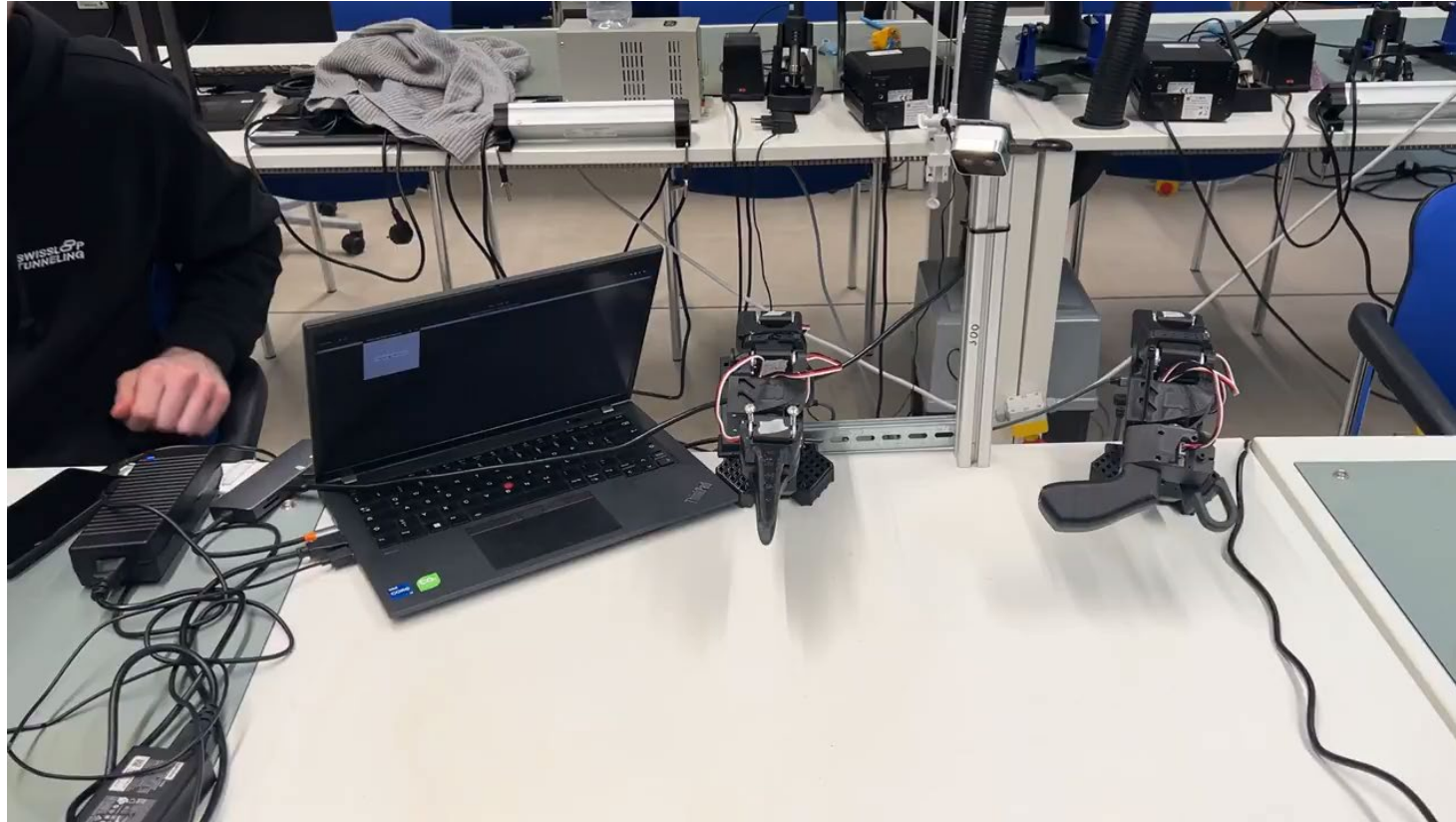
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Example: Folding Towel



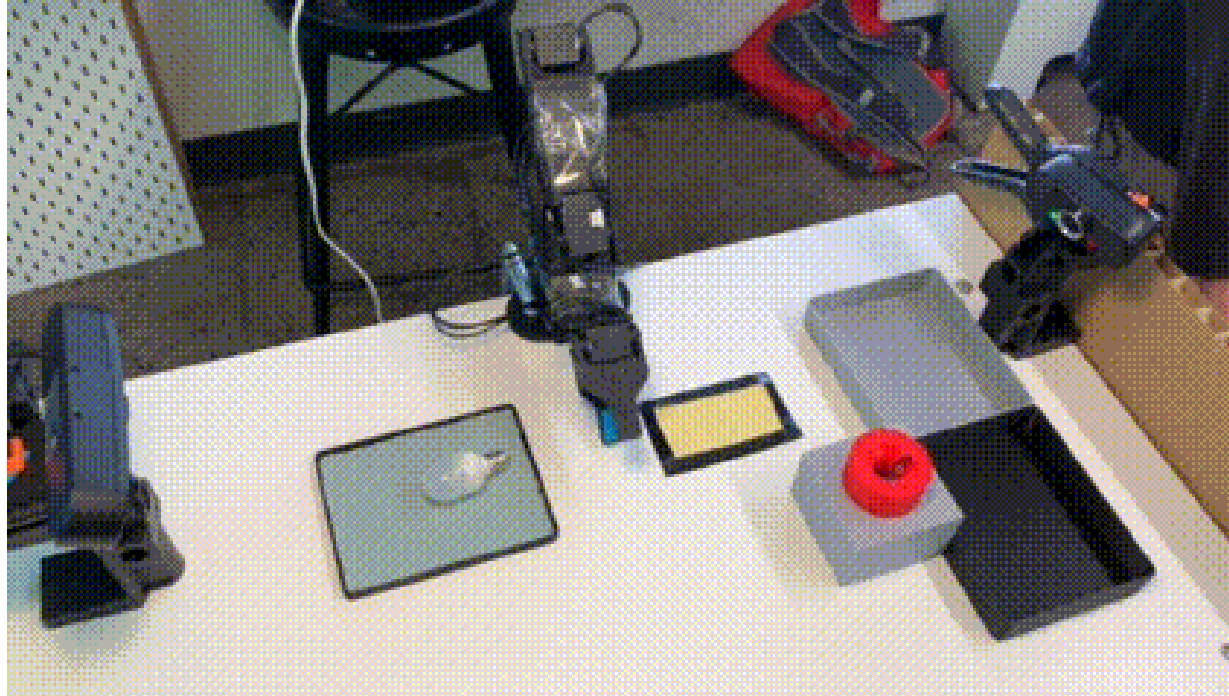
<https://x.com/cmcgartoll/status/1909803805505368535>

Example: Cleaning Table



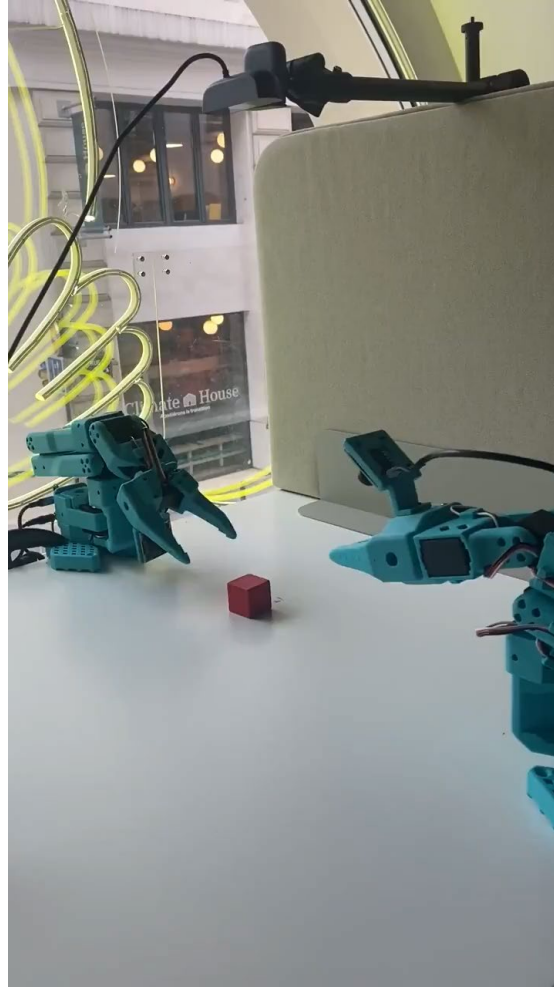
<https://x.com/JannikGrothusen/status/1852790503823057073>

Example: Peg-in-Hole



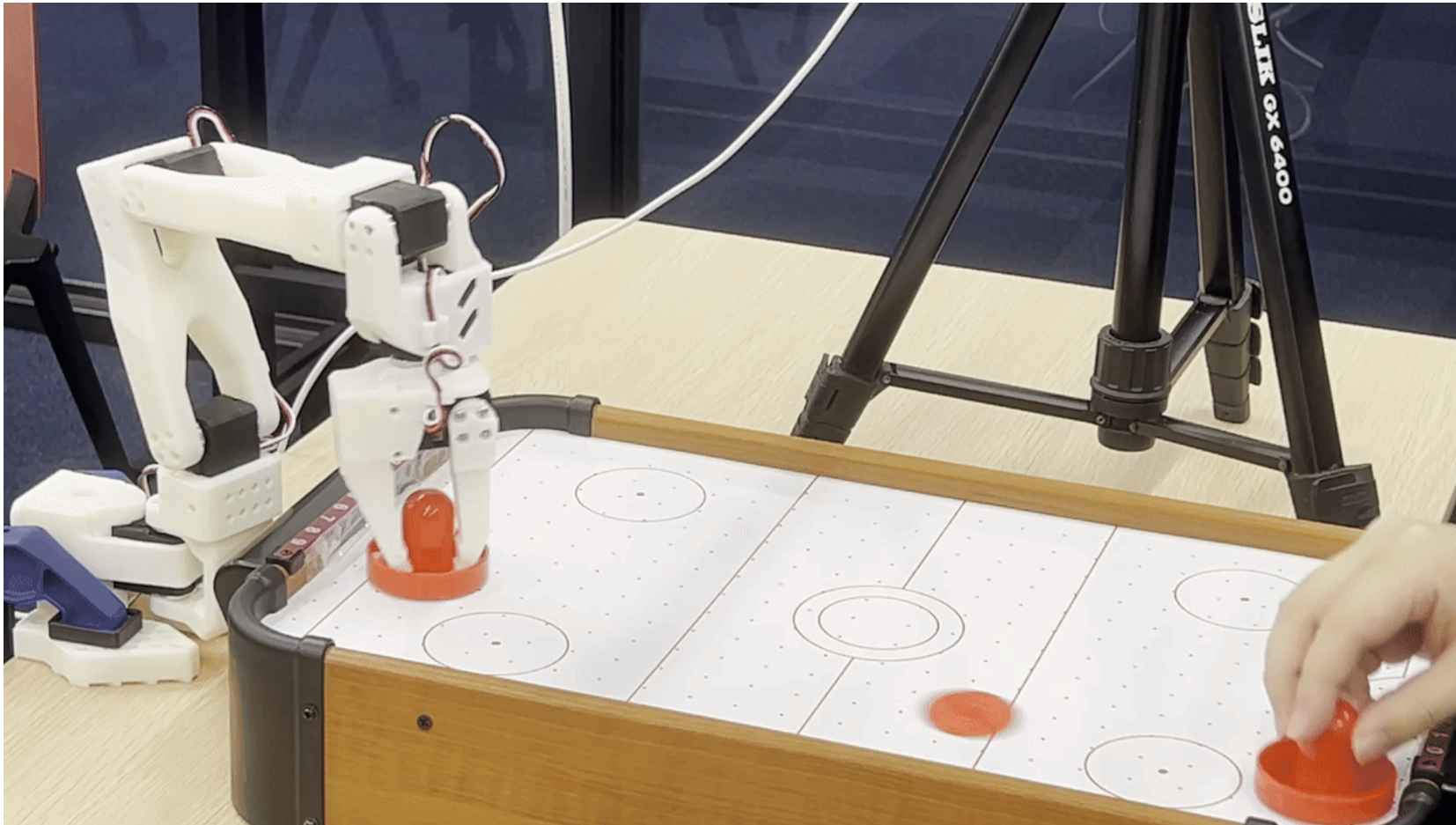
<https://github.com/SimpleAutomationOrg/SimpleAutomation>

Example: Robot-to-Robot Handover



<https://x.com/LeRobotHF/status/1945818023291482563>

Example: Hockey



<https://huggingface.co/datasets/LeRobot-worldwide-hackathon/134-npaka-studio-air-hockey>

Example: Teleoperation



<https://x.com/AgilexRobotics/status/1963886533381157333>

Example: ChessBot



<https://x.com/ahadj0/status/1934432727635484776>

Example: Mobile Manipulation



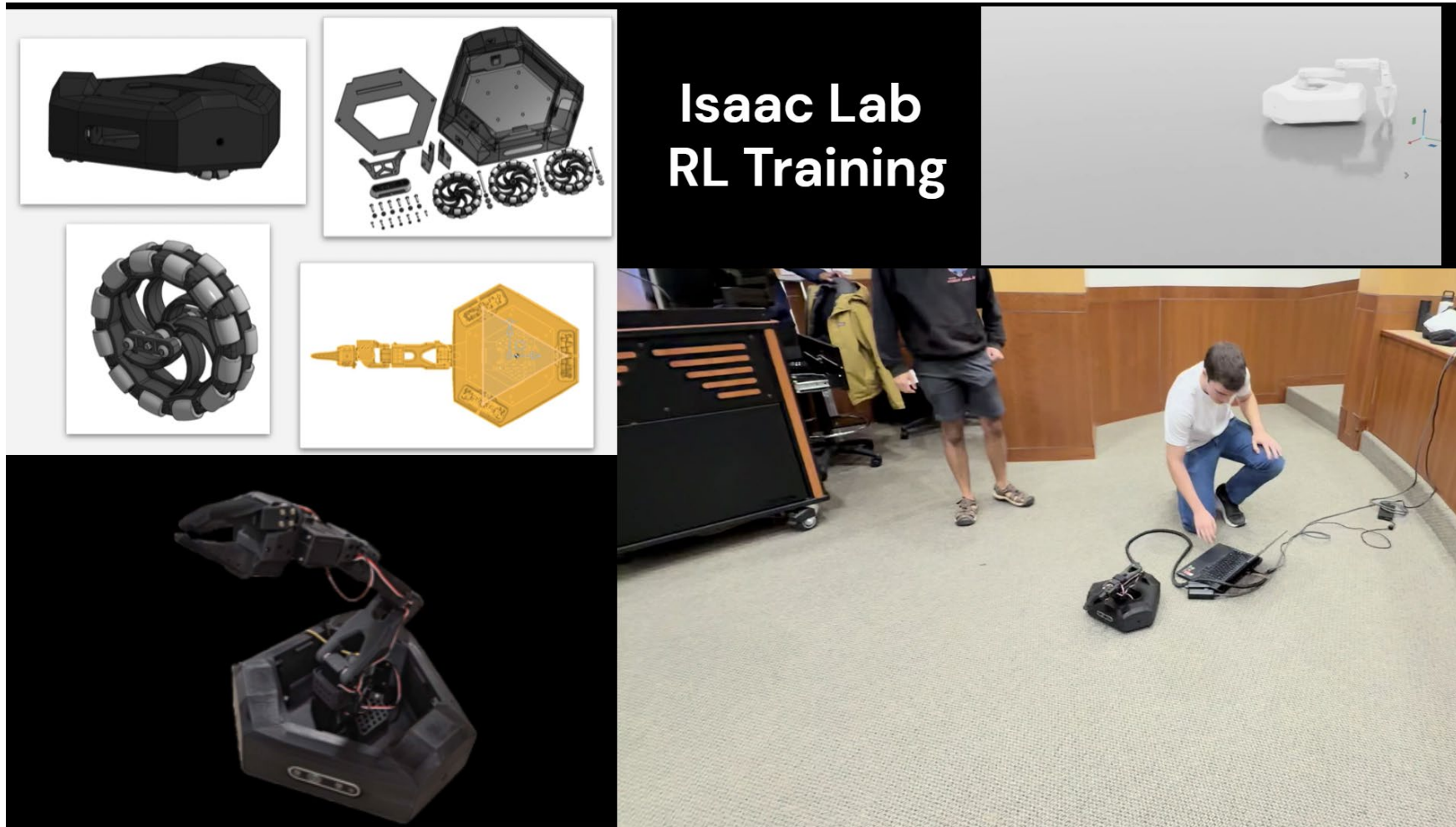
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Example from Last Year Fall 2025



<https://labs.utdallas.edu/irvl/courses/fall-2025-cs-6341-robotics/>

Example from Last Year Fall 2025



<https://labs.utdallas.edu/irvl/courses/fall-2025-cs-6341-robotics/>

Paradigm 1: Model-based Manipulation

Perception



Planning

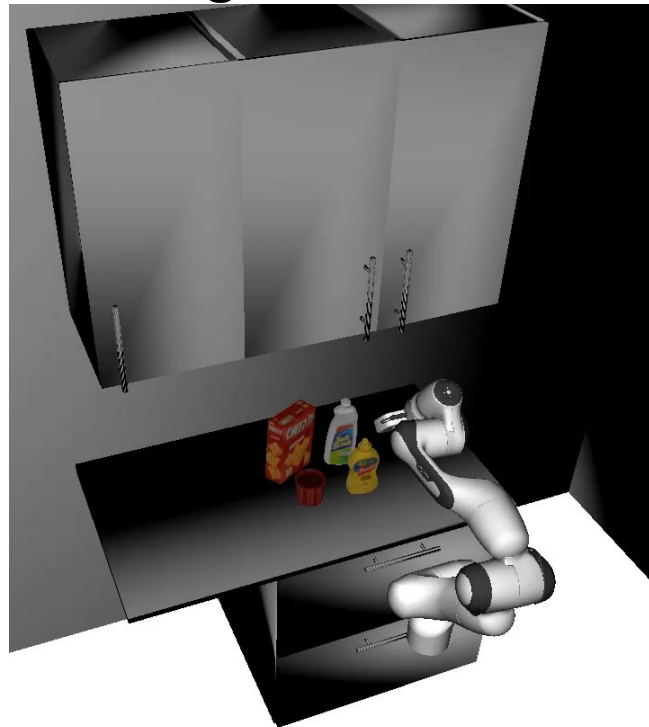


Control

Sensed image



Planning scene



Real world execution

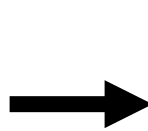


Paradigm 2: Learning-based Manipulation

Teleoperation

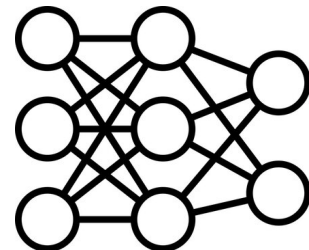


Collect Demonstrations



(state, action)

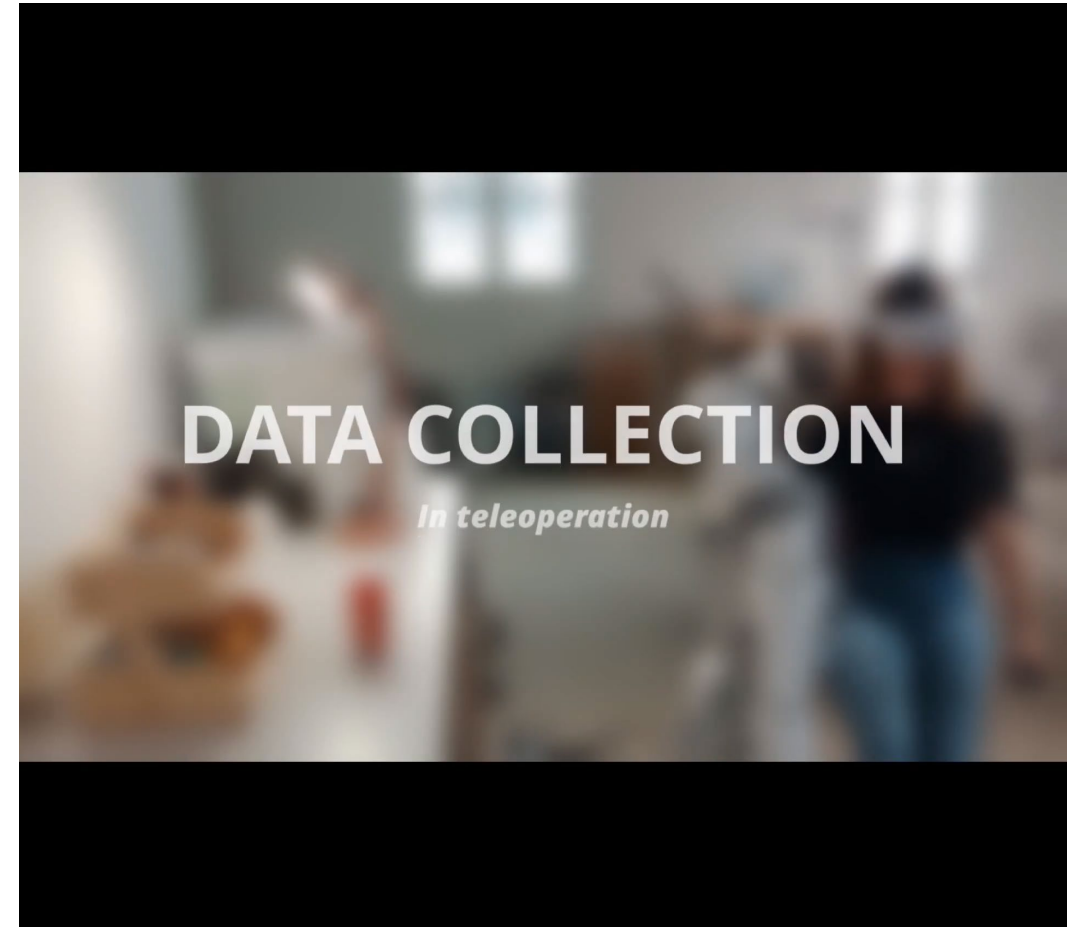
A Dataset of State-Action Pairs



Train a Policy Network

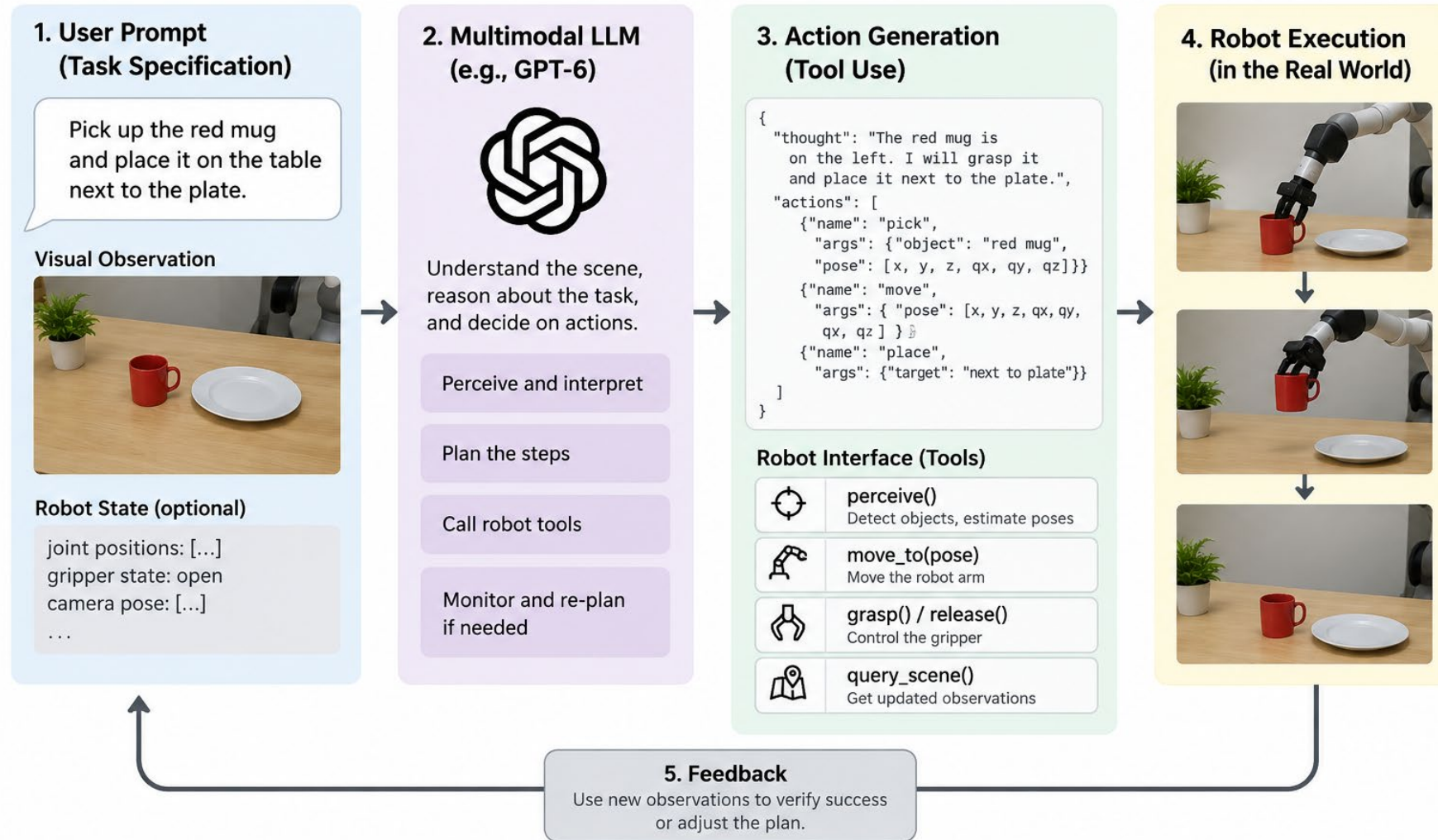


Deploy the Policy Network



<https://x.com/LeRobotHF/status/1963893227708133763>

Paradigm 3: Prompt-based Manipulation



Examples with GPT 6 Astra



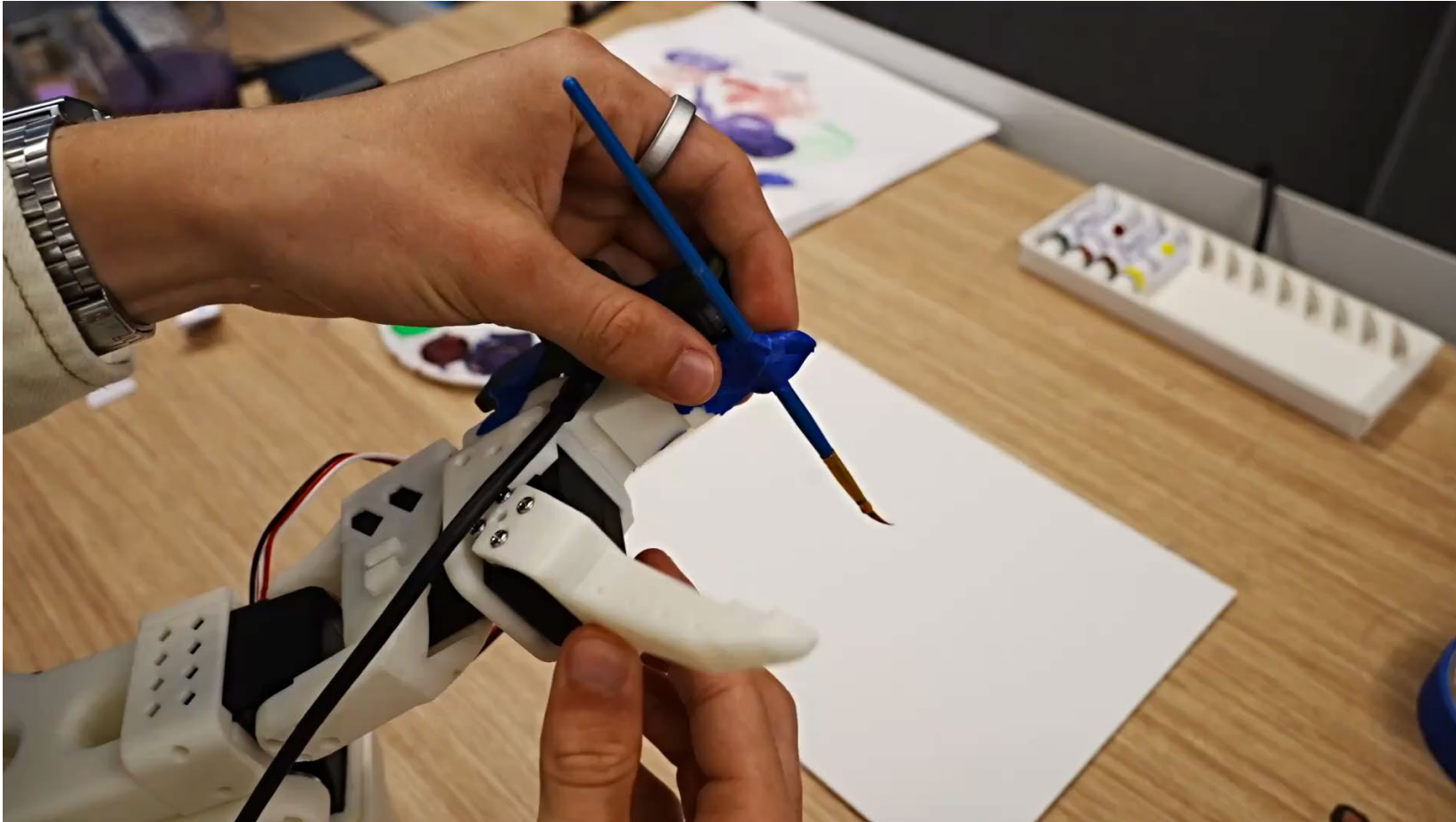
https://x.com/chooi_jeq/status/2096064315115839904?s=20

Examples with GPT 6 Astra

- Prompt: “You are controlling a real robot embodiment named 'yam_arms' through tool calls. Each observation message gives you the current proprioceptive state and camera images. Work toward the user's goal in small, deliberate motions; re-check the observation after every motion. Every move tool call must include a note: in one or two sentences, say what you observe in the current observation and why you chose this motion. The user is watching these notes to see what you see and what you decide, so write them for a human reader. Safety approvers clamp out-of-bounds and too-fast actions below you. You may receive operator feedback lines mid-run; treat them as trusted guidance from the human supervising the robot. Respond with exactly one tool call per turn. When the goal is achieved call done; if it cannot be achieved call give_up. Note what you are learning about this rig and task as you go: done and give_up will ask what you wish you had known from the start. You have a budget of 20 LLM calls for the whole trial.
- Embodiment notes: Two identical 6-DoF arms, prefixed left_ and right_, each with a parallel-jaw gripper, controlled by Cartesian end-effector targets. Each arm's targets are in that arm's own base frame: +x points forward out of the base, +y left, +z up; how the two bases are mounted relative to each other depends on the rig. left_x / right_x, left_y / right_y, left_z / right_z: grasp-point position in meters in the arm's base frame (the grasp point sits between the fingertips). left_yaw / right_yaw: tool rotation in radians about vertical, relative to the trial's start orientation; 0 keeps the start orientation and positive turns counterclockwise seen from above. left_pitch / right_pitch, left_roll / right_roll: tool tilt in radians, also relative to the trial's start orientation. Positive pitch tips the tool forward (+x at yaw 0), positive roll toward the arm's left (+y at yaw 0). An axis whose configured bounds are equal (typically 0) is pinned: commands on it are clamped to that value and it cannot be actuated. left_gripper / right_gripper: 0 is fully closed, 1 is fully open (about 9.5 cm between the jaws).
- Proportions: upper arm 0.26 m, forearm 0.25 m, wrist to grasp point 0.25 m when straight; reach from the shoulder about 0.76 m. An inverse-kinematics layer converts targets into joint motion; unreachable or awkward targets may be tracked slowly or held, so prefer modest steps and re-check the observation after each motion.
- Joint effort: observation.state["joint_eff"] is per-joint estimated torque in N·m, sign-corrected, with the same 14-slot layout as joint_pos but raw effort in the gripper slots. The values include gravity load, so compare against a moving baseline. Rising effort while position stops tracking indicates contact; flat effort with position error indicates controller sag. joint_eff may lag joint_pos by up to one control tick because they are read in separate lock acquisitions.”
- User: “Goal: Pick up the round blue puzzle piece by the knob at its center and place it into the matching circular groove in the board. Use your right arm only.”

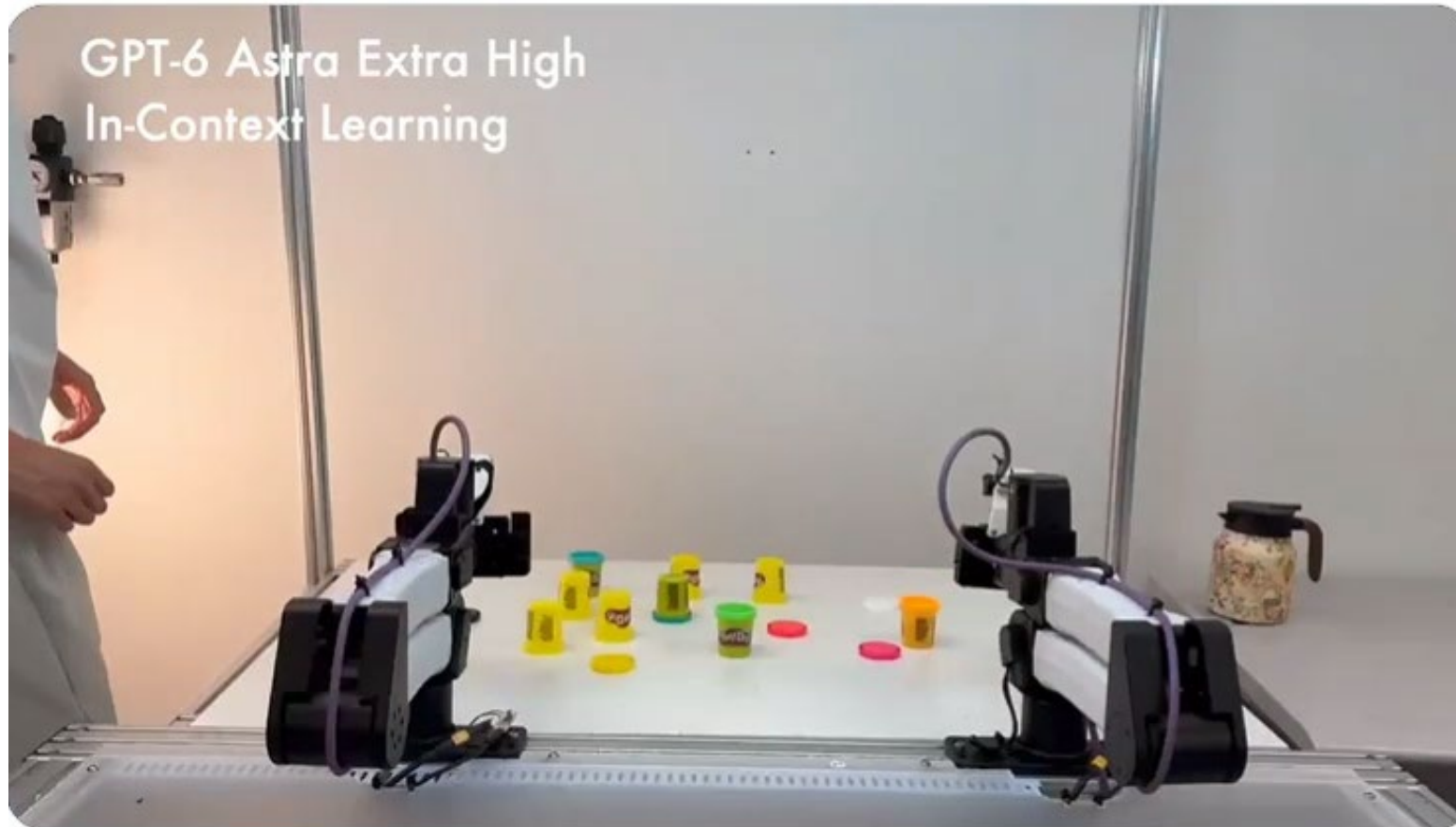
https://anthropic.robocurve.org/fable-5.1/runs/transcripts/rig-3_f3f30ea2

Examples with GPT 6 Astra



<https://x.com/cdngdev/status/2097339677128982873?s=20>

Examples with GPT 6 Astra



<https://x.com/wenlixiao/status/2097801944119349455?s=20>

Resources

- LeRobot homepage: <https://github.com/huggingface/lerobot>
- Pretrained models and datasets: <https://huggingface.co/lerobot>
- SmolVLA: Efficient Vision-Language-Action Model trained on Lerobot Community Data: <https://huggingface.co/blog/smolvla>
- SO101 arm with ROS2: https://github.com/Pavankv92/lerobot_ws
- Harness <https://github.com/robocurve/inspect-robots>
- CS6341 Fall 2025 <https://labs.utdallas.edu/irvl/courses/fall-2025-cs-6341-robotics/>
- Search more online...

Propose Your Projects

- Which track is your project?
 - Research-oriented? Application-oriented? Implementation-oriented?
- What task you want the SO101 arm to do?
 - **Novelty**
- Will you do model-based or learning-based or prompt-based?

Project Proposal will be due next Friday

- Think of what to work on
- Finish the SO101 arm calibration
- Contact the TA for teleoperation testing (TA office hour Monday, 2PM-4PM, in ECSS 4.222)
- Take a mp4 video of the teleoperation for submission

Demo time and Discussion



TA: Qifan Zhang